



Separating policy burdens: quantifying the economic benefits and mechanisms in China's state-owned enterprises

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ABSTRACT

This study explores the economic benefits and mechanisms of removing policy burdens in Chinese state-owned enterprises (SOEs), focusing on the reforms related to the separation of public services such as 'Heating, Power, Water, and Estate' (HPWE), which were previously provided to employees by SOEs for a fee. Utilizing this reform as a quasi-natural experiment, the research demonstrates significant performance improvements in SOEs following the HPWE separation, as confirmed by robustness tests. Mechanism analysis indicates that the HPWE separation enhances firm performance by reducing redundant personnel, increasing labour productivity, and lowering agency costs. These effects are particularly pronounced in labour-intensive SOEs with a greater separation between ownership and control. This article offers recommendations for improving the performance of SOEs from the perspective of policy burdens separation, thereby contributing a new perspective to the literature on SOE reforms.

KEYWORDS

State-owned enterprises; policy burdens; heating, power, water and estate; firm performance

JEL CLASSIFICATION

M41; P31; G38

1. Introduction

SOEs play a crucial role in China's socialist market economy, with ongoing reforms targeting improved operational efficiency. However, SOEs still encounter significant inefficiencies stemming from policy burdens (Lin 2021). These burdens compel SOEs to divert considerable resources towards non-core activities, adversely affecting their financial health. Policymakers attempt to alleviate these losses through government subsidies, yet asymmetric information frequently results in misreporting and soft budget constraints (Lin and Li 2008; Lin, Cai, and Li 1998). Moreover, these burdens influence the turnover sensitivity of chairpersons and amplify executive perks (Jian et al. 2020). The contradiction between the critical role and low efficiency of SOEs is a global issue. Trihatmoko and Susilo (2023) examined the Indonesia Raya Incorporated (IRI) framework, highlighting the role of governments and SOEs in managing essential resources (Trihatmoko and Susilo 2024). Although strategically important, SOEs often face multiple, conflicting objectives (Mai and Casady 2023), and institutional pressures drive them to assume extensive social responsibilities (de Pilla, Peci, and Leite 2024). These non-commercial obligations tend to hinder market growth, inflate

agency costs (Apriliyanti, Dieleman, and Randøy 2024), and decrease overall productivity. Similar inefficiencies have been observed globally, across both developing regions like Eastern Europe (Matuszak and Szarzec 2019) and Vietnam (Phuong, Nguyen, and Vu 2020), as well as developed nations such as Italy (Cardinale and Belotti 2022), Singapore (Paiva-Silva 2022), and Norway (Goldeng, Grünfeld, and Benito 2008).

It is widely acknowledged that policy burdens impede the operational efficiency of SOEs (Hartwell and Korovkin 2024), and the literature extensively discusses privatization as a method to improve SOEs efficiency (Arnold 2022; Zhang, Yu, and Chen 2020; Zhao and Mao 2023). However, the specific mechanisms by which policy burdens affect SOEs performance remain underexplored. Lin and Li (2008) provides a theoretical framework for understanding the negative impacts of policy burdens on SOEs, but it lacks empirical support from large-sample studies. Additionally, empirical research on SOEs policy burdens, such as that by G. Liao et al. (2009), which uses excess employees as a proxy, is limited by endogeneity issues, including reverse causality and omitted variables.

To bridge this gap, this study examines China's policy burdens reform for SOEs, specifically analysing the impact of removing social obligations on their performance. During the era of China's planned economy, insufficient local public services compelled SOEs to develop unrelated infrastructure to provide public services, a phenomenon known as 'running society'. Among the various social functions, providing 'Heating, Power, Water, and Estate' (HPWE) services for employee residential areas represents the heaviest policy burden. Although employees fund these services, managing these operations remains inefficient for the enterprises. As the market economy evolved and urban public service levels improved, the social functions became increasingly burdensome, hindering normal operations and fair market competition. To address these issues, China initiated the separation of these functions from SOEs. Since 2012, the State-owned Assets Supervision and Administration Commission (SASAC) has spearheaded the HPWE reform, starting with a pilot programme in Heilongjiang Province before expanding to Henan, Hunan, and Chongqing. By 2018, SASAC mandated the completion of HPWE separation, significantly impacting numerous enterprises and employees.

This study investigates the effects of the HPWE separation reform on China's A-listed SOEs from 2009 to 2021. Employing a staggered Difference-in-Differences (DID) approach, the study shows that HPWE separation significantly boosts SOE performance, as validated by various robustness tests, such as parallel trends, placebo, and control group replacement tests. Mechanism analysis indicates that HPWE separation primarily enhances performance by reducing redundant personnel, increasing labour productivity, and decreasing agency costs. Furthermore, heterogeneity analysis reveals that performance improvements are more substantial in SOEs with higher labour intensity and greater separation between ownership and control.

The paper makes three main contributions. First, it explores the economic consequences of the separation of 'HPWE', expanding the literature on the policy burdens of SOEs. While policy burdens are a well-documented issue for SOEs, the existing literature is limited, with much of the research on SOE reform focusing on privatization

(Arnold 2022; Brown, Earle, and Telegdy 2016; Rakhman 2018) or mixed-ownership reform (Zhang, Yu, and Chen 2020; Zhao and Mao 2023). Second, this paper quantitatively evaluates the economic benefits and mechanisms of the 'HPWE' separation reform, with a particular emphasis on corporate labour mechanisms. In contrast, prior studies primarily examine how policy burdens impact corporate performance through credit allocation (Ye et al. 2021) or tax incentives (F. Liao, Sun, and Xu 2023). Finally, this study utilizes the pilot promotion of social function separation reform, an exogenous top-down policy initiated by the state, as a quasi-natural experiment, addressing the endogeneity issues that have been prevalent in previous studies on SOE policy burdens (G. Liao et al. 2009; Ye et al. 2021).

The paper is structured as follows: [Section II](#) establishes the theoretical framework and hypotheses; [Section III](#) outlines the research design and data collection methods; [Section IV](#) presents the main empirical findings and robustness tests; [Section V](#) conducts additional mechanism and cross-sectional tests; and finally, [Section VI](#) discusses policy implications and concludes the paper.

II. Theoretical analysis and hypothesis formulation

SOEs' policy burden separation and performance enhancement

In socialist countries, misalignment of objectives between the government and SOEs is a significant issue. Iwasaki, Ma, and Mizobata (2022) analyse EU countries in Eastern Europe, Russia, and China, finding that the lack of regulatory and incentive mechanisms leads to divergent goals between management and shareholders, negatively affecting corporate performance, particularly in resource-intensive sectors. Phuong, Nguyen, and Vu (2020) emphasize that political and bureaucratic interventions in Vietnam prioritize political interests over operational efficiency. Matuszak and Szarzec (2019) further note that SOEs in post-socialist Eastern Europe underperform on various financial metrics compared to private enterprises, facing high employee costs as well.

In China, the social objectives imposed on SOEs create a significant policy burden due to extensive government intervention in business decisions (Lin and Tan 1999). Chinese SOEs initially provided social welfare, but ongoing political pressures and profit deficiencies hinder investment efficiency, despite corporatization reforms (Bo, Li, and Toolsema 2009). Chan (2009) observes that SOEs often prioritize public management over profit, consuming economic resources and diminishing operational performance and stock returns while contributing to redundancy issues (Lin 2021). Empirical studies indicate that these policy burdens hinder investment efficiency, making their reduction essential for enhancing economic efficiency and market competitiveness (F. Liao, Sun, and Xu 2023; Ye et al. 2021).

The separation of 'HPWE' is a key strategy for alleviating the policy burden on SOEs, enabling them to focus on core operations and reduce the negative effects of unrelated diversification (Singh, Nejadmalayeri, and Mathur 2007). This challenge complicates SOEs' ability to concentrate on primary activities amid advancing urban public services. The benefits of 'HPWE' separation include optimal resource allocation for greater investment in profit-generating activities, which enhances efficiency (Kang and Kim 2012); alleviation of soft budget constraints, as Lin and Li (2008) argue that privatization alone cannot resolve these issues without removing policy burdens; and reduced social responsibilities, which optimize personnel allocation and lower labour costs. Furthermore, minimizing government intervention fosters a freer market, enhancing decision-making autonomy and aligning executive incentives with corporate performance, thus promoting long-term development and profitability (G. Liao et al. 2009). Therefore, we propose Hypothesis 1 (H1) as follows:

H1: The separation of 'HPWE' reform will effectively enhance the performance of SOEs.

SOEs' policy burden separation mechanism and performance impact

Reduction of redundant personnel and labour cost. SOEs often face excessive employment challenges

due to their obligation to maintain social stability, leading to rising labour costs (Gu, Tang, and Wu 2020). This issue is also evident in Eastern European countries, where SOEs struggle with low capital utilization rates and high labour costs (Matuszak and Szarzec 2019). Despite government efforts to alleviate this burden through measures such as tax incentives and subsidies (Lin and Tan 1999), the effectiveness of these interventions has been limited, and excessive employment continues to significantly impact corporate performance (G. Liao et al. 2009). Particularly in the case of HPWE, which are predominantly non-profit in nature, there tends to be a misallocation and waste of labour resources. Reforming HPWE through asset transfers and employee redeployment is expected to reduce redundant labour, lower costs, and ultimately enhance corporate performance. Thus, we propose Hypothesis 2:

H2: Separating HPWE will optimize labour allocation, reduce redundancy, and enhance SOE performance.

Government interference in SOEs is often driven by political purposes rather than aimed at improving corporate efficiency (Phuong, Nguyen, and Vu 2020), which can lead to excessive employment and resource waste. In contrast, the privatization of SOEs has been shown to significantly boost productivity (Brown, Earle, and Telegdy 2016). Similarly, HPWE reforms should positively impact labour efficiency, as they typically consume substantial subsidies and occupy personnel quotas, straining SOEs' economic resources. Post-reform improvements in infrastructure and services can enhance employees' living standards and motivation through better benefits and work environments. Additionally, releasing personnel quotas and resources will enable SOEs to attract higher-quality employees, further enhancing productivity through effective team management and skill training. Therefore, we propose Hypothesis 3:

H3: Separating HPWE will enhance labour productivity and optimize resource allocation, thereby improving SOE performance.

Clarification of management responsibilities and reduction of agency costs. Excessive policy burdens, such as HPWE, often obscure the management responsibilities of SOE executives. This issue of conflicting political objectives and corporate strategies is also evident in Russian state-owned enterprises (Libman, Stone, and Vinokurov 2022). As a result, the focus on corporate performance during government evaluations diminishes, leading managers to engage in riskier behaviours to evade accountability, such as overinvestment and assuming heavy tax burdens (Bradshaw, Liao, and Ma 2019; Cong et al. 2019). Reforming HPWE can address these challenges by reducing non-core management responsibilities, allowing executives to concentrate on essential business operations. This shift clarifies responsibility boundaries and enhances the alignment between corporate performance and compensation systems, ultimately reducing agency costs. Therefore, we propose the following hypothesis: H4:

H4: Separating HPWE will improve SOE performance by clarifying management responsibilities and reducing agency problems.

III. Methodology

Model specification and variable definition

To investigate the impact of separating the HPWE on firm's performance, we adopt the methods proposed by Bertrand and Mullainathan (2003) and Beck, Levine, and Levkov (2010). Thus, the regression model is set up as follows:

$$Froa_{i,t+1} = \beta_0 + \beta_1 DID_{i,t} + \Sigma Controls + \Sigma Year + \delta_i + \varepsilon_{i,t} \quad (1)$$

In model (1), the subscripts i and t represent firm and year respectively. The dependent variable $Froa_{i,t+1}$ is the business performance of firm i in year $t + 1$, represented by the return on total assets. The independent variable $DID_{i,t}$ indicates whether SOEs have transferred the HPWE. If SOEs transfer the HPWE in a certain year, DID equals 1 in that year and the following years, and 0 otherwise. Model (1)

also adds some control variables that may simultaneously affect both the reform of HPWE and the return on total assets. These control variables include: firm size (*Size*), capital structure (*Lev*), profitability (*Roa*), cash holdings (*Cash*), inventory (*Inventory*), fixed assets (*Ppe*), operating cashflow (*Cfo*), investment expenditure (*Capexp*), R&D expenditure (*R&D*), investment opportunity (*TobinQ*), state-owned enterprise level (*LocalSOE*), ownership concentration (*Top10*), duality (*Dual*), ownership balance (*Stkba*), board independence (*Indep*). The specific definitions of all variables can be found in Table 1. δ_i and $Year$ denote the firm fixed effect and the time fixed effect, respectively. $\varepsilon_{i,t}$ is the random disturbance term. We cluster all the regression standard errors at firm level. Parameter β_1 indicates the impact of separating the HPWE on firm's performance. Based on Hypothesis1, β_1 should be significantly positive.

Sample selection

The HPWE reform, initiated in 2012 for central enterprises in Heilongjiang, expanded to other regions by 2014 and nationwide by 2016. Despite the 'Guiding Opinions' establishing a deadline for completion by 2018, actual progress varied due to regional complexities. To accurately track HPWE transfer dates, we manually collected data from sources like the China Economic Information Service and SASAC website. Huaneng Power International led the reform, with over 60% of SOEs implementing it by 2018. Using A-share state-owned listed firms from 2009 to 2021, excluding financial and ST samples, we constructed a final sample of 12,135 firm-year observations from the China Stock Market & Accounting Research Database (CSMAR).

IV. Empirical results

Descriptive statistics

Table 2 presents the descriptive statistics of all variables. The mean value of *Froa* is 0.0281, and the standard deviation is 0.0612, indicating significant variation in the firm performance of SOEs. In addition, we calculated the proportion of SOEs that underwent the HPWE reform during the sample period indicated by the variable *Treat*. The mean value of

Table 1. Variable definitions.

Variable	Definition
<i>Froa</i>	Net profit divided by total assets in $t+1$
<i>DID</i>	Equals 1 after this firm transfers the related HPWE assets, otherwise 0.
<i>Treat</i>	Equals 1 if the firm transfers the related HPWE assets within the sample period, otherwise 0.
<i>Size</i>	The natural logarithm of the firm's total assets
<i>Lev</i>	Total liabilities divided by total assets
<i>Roa</i>	Net profit divided by total assets
<i>Cash</i>	Cash divided by total assets
<i>Inventory</i>	Inventory divided by total assets
<i>Ppe</i>	Fixed assets divided by total assets
<i>Cfo</i>	Operating cash flow divided by total assets
<i>Capexp</i>	The cash paid for the purchase and construction of fixed assets, intangible assets and other long-term assets divided by the total assets
<i>Apply</i>	The natural number of patent applications
<i>TobinQ</i>	The sum of the market value of the stock and the face value of the debt divided by the face value of the total assets
<i>LocalSOE</i>	Equals 1 for local state-owned enterprises and 0 otherwise
<i>Top10</i>	The proportion of shares held by the top ten shareholders
<i>Dual</i>	Equals 1 when chairman and the general manager CEO are the same, otherwise 0
<i>Stkba</i>	The sum of the shareholding ratio of the second to tenth shareholders divided by the shareholding ratio of the largest shareholder
<i>Indep</i>	The proportion of the number of independent directors in the total number of directors of an enterprise

This table presents main variable definitions used in our analyses.

Treat is 0.2246, indicating that approximately 22.5% of SOEs implemented the HPWE reform during the sample period. The mean value of *DID* is 0.0632, indicating that 6.32% of the samples are influenced by the HPWE reform. The descriptive statistics of other variables are shown in Table 2.

Baseline regression

Table 3 presents the regression results on whether the HPWE reform affects the firm performance of SOEs. The dependent variable in columns (1) and (2) is the return on total assets (*Froa*) of firm i in the next period. From the regression results in columns (1) and (2), it is evident that even after controlling for other variables, the coefficient of *DID* is significantly positive at the 1% level,

indicating that the HPWE reform can help improve the firm performance of SOEs. The empirical results support *Hypothesis 1*. In terms of economic significance, compared with other SOEs, the return on total assets of the SOEs that underwent the HPWE reform has increased by approximately 29% after the reform, which is significant.

Robustness test

Parallel trend test

An important prerequisite for using the difference-in-differences model is that the control group and the treatment group must satisfy the parallel trend assumption of the model. Therefore, according to Bertrand and Mullainathan (2003), we use the following model for testing:

Table 2. Descriptive statistics.

Variable	N	Mean	SD	p1	p25	p50	p75	p99
<i>Froa</i>	12135	0.0281	0.0607	-0.2064	0.0092	0.0277	0.0527	0.1819
<i>DID</i>	12135	0.0632	0.2433	0	0	0	0	1
<i>Treat</i>	12135	0.2246	0.4174	0	0	0	0	1
<i>Size</i>	12135	22.668	1.4647	19.824	21.670	22.507	23.552	26.771
<i>Lev</i>	12135	0.5139	0.2104	0.07520	0.3566	0.5203	0.6699	0.9785
<i>Roa</i>	12135	0.0283	0.0612	-0.2207	0.0096	0.0281	0.0535	0.1789
<i>Cash</i>	12135	0.1676	0.1221	0.0129	0.0833	0.1362	0.2176	0.5880
<i>Inventory</i>	12135	0.1576	0.1578	0.0001	0.0492	0.1168	0.2044	0.7449
<i>Ppe</i>	12135	0.2612	0.1969	0.0019	0.0987	0.2191	0.3932	0.7725
<i>Cfo</i>	12135	0.0457	0.0716	-0.1635	0.0065	0.0457	0.0875	0.2421
<i>Capexp</i>	12135	0.0446	0.0448	0.0002	0.0124	0.0308	0.0621	0.2142
<i>Apply</i>	12135	2.4881	2.0330	0	0	2.5649	3.9890	7.4061
<i>TobinQ</i>	12135	2.0644	1.5381	0.8405	1.1581	1.5606	2.3535	9.0204
<i>LocalSOE</i>	12135	0.6651	0.4720	0	0	1	1	1
<i>Top10</i>	12135	0.5742	0.1594	0.2329	0.4593	0.5746	0.6882	0.9203
<i>Dual</i>	12135	0.1016	0.3021	0	0	0	0	1
<i>Stkba</i>	12135	0.6399	0.6256	0.0305	0.1870	0.4304	0.8956	2.8264
<i>Indep</i>	12135	0.3703	0.0583	0.3000	0.3333	0.3333	0.3846	0.5714

This table reports descriptive statistics of the main variables used in our analyses.

Table 3. The effect of the HPWE reform on firm performance.

	(1) <i>Froa</i>	(2) <i>Froa</i>
<i>DID</i>	0.0108*** (3.7889)	0.0083*** (3.3207)
<i>Size</i>		-0.0090*** (-4.6657)
<i>Lev</i>		0.0117 (1.3921)
<i>Roa</i>		0.0884*** (3.3304)
<i>Cash</i>		0.0289*** (3.2255)
<i>Inventory</i>		-0.0020 (-0.1971)
<i>Ppe</i>		-0.0096 (-0.9919)
<i>Cfo</i>		0.0894*** (7.8885)
<i>Capexp</i>		0.0164 (1.0172)
<i>Apply</i>		0.0012* (1.9482)
<i>TobinQ</i>		0.0075*** (7.9737)
<i>LocalSOE</i>		-0.0077 (-1.4126)
<i>Top10</i>		0.0531*** (5.0626)
<i>Dual</i>		0.0040* (1.8114)
<i>Stkba</i>		-0.0021 (-0.9114)
<i>Indep</i>		-0.0010 (-0.0734)
<i>Constant</i>	0.0274*** (152.3419)	0.1735*** (4.1508)
Year FE	Yes	Yes
Firm FE	Yes	Yes
Observations	12135	12135
Adj.R ²	0.4218	0.4640

This table reports the estimation results of regressions that examine the impact of the HPWE reform in SOEs on corporate performance. The dependent variables is *Froa* of firm *i* in the next period. The independent variable is *DID*, which represents whether SOEs have handed over the HPWE. All variables are defined in Table 1. Across all specifications, we include year and firm fixed effects. The significance is * $p < 0.05$ ** $p < 0.01$ *** $p < 0.001$. Robust standard errors are in parentheses and are clustered at the firm level (Unless otherwise specified, the same fixed effects and standard errors are used in the subsequent regressions in this article).

$$Froa_{i,t+1} = \beta_0 + \sum \beta_s Ds + \sum Controls + \sum Year + \delta_i + \varepsilon_{i,t} \quad (2)$$

In Model (2), the independent variable (*DID*) from Model (1) is replaced with dummy variables representing specific time windows: three years before the reform (*D-3*), two years before (*D-2*), one year before (*D-1*), the reform year (*D0*), and the first (*D1*), second (*D2*), third (*D3*), and fourth and beyond years after the reform (*D4+*). The remaining sample is the baseline group. We regress Model

(2) to obtain coefficients and 95% confidence intervals for each time window, as shown in Figure 1. Figure 1 reveals that before the HPWE reform, coefficients are insignificant; from the first year after, coefficients become significantly positive and show a gradual upward trend. The insignificant reform year coefficient may result from initial maintenance costs related to HPWE. These findings confirm the parallel trend hypothesis between the treatment and control groups.

Placebo test

To address potential endogeneity from measurement errors and omitted variables, we use the placebo testing method from Li, Lu, and Wang (2016). We randomly assign treatment and control groups from the original samples 1,000 times, obtaining coefficients and p-values for each regression. The distribution of these virtual coefficients and p-values, shown in Figure 2, follows a normal distribution, with most coefficients near zero and smaller than those from the actual regression results in Table 3. This suggests that any endogeneity issues from measurement errors and omitted variables in this study are likely minor.

Adjust research sample

We manually collected data to determine when each SOE implemented the HPWE reform. However, some SOEs did not specify the exact timing of transferring HPWE-related assets. Following the principle of ‘transfer first, then reform’, we used the disclosure date of HPWE-related information as the transfer date, setting the independent variable (*DID*) accordingly. This may lead to delayed identification of transfer times for some SOEs, introducing measurement errors into the *DID* variable. Despite this potential bias against the results showing improved SOE performance post-HPWE reform, the *DID* coefficient remains significant in the main regression results, even without this factor.

To validate our identification method and accurately estimate the HPWE reform effect, we first exclude samples from the reform year and conduct regressions using model (1), as shown in column (1) of Table 4. Then, we remove samples from the year before and after the reform and perform regressions using model (1), with results presented in column (2) of Table 4. The coefficients of *DID* in both

columns (1) and (2) of Table 4 are significantly positive, consistent with the main findings. Moreover, after excluding the reform year samples, the coefficients of *DID* are higher than those in the main results. These findings indicate that after accounting for timing errors, the impact of the HPWE reform on business performance increases, affirming the robustness of our empirical results.

Add the control variable

To address potential regional omitted variable bias, model (1) controls for regional public service expenditure (*Public*), marketization degree (*Market*), and provincial fixed effects. Public expenditure is calculated as general public service spending divided by budget revenue, while marketization degree uses a marketization index. Results in Table 5 (1) show

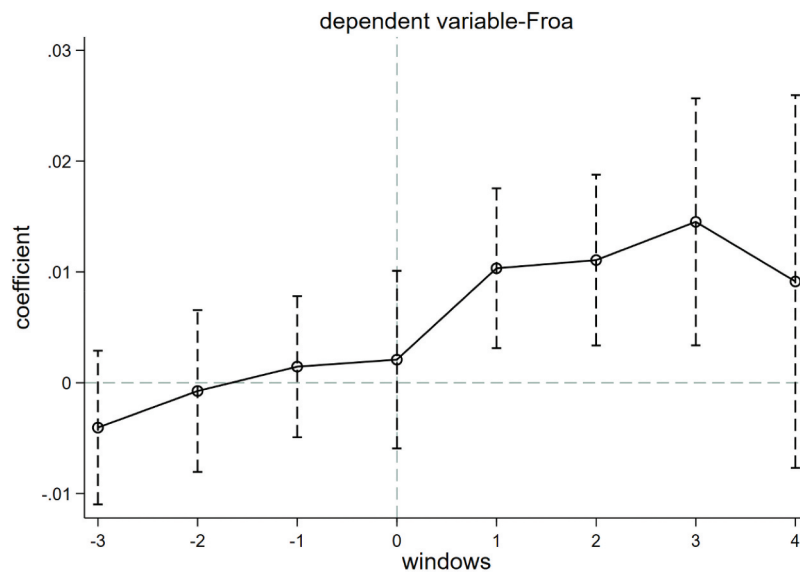


Figure 1. Parallel trend test. This figure graphs the parallel trend of the HPWE reform in SOEs on corporate performance. The dynamic effect is represented by connected dots; dashed 'arms' around the dots represent 95% confidence intervals, adjusted for firm-level clustering. The vertical solid line indicates the year in which the companies immediately complete the HPWE reform.

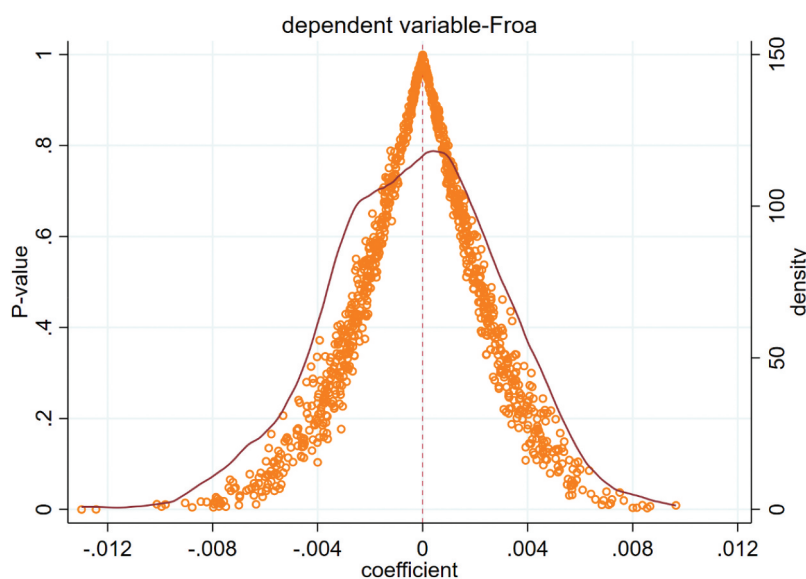


Figure 2. Placebo test. This figure plots the placebo test of the HPWE reform in SOEs on corporate performance. We randomly assigned the treatment group and the control group 1000 times to regress and then obtain the corresponding coefficient and p-value. The yellow line indicates the distribution of coefficients and the yellow hollow dot indicates the corresponding p-values.

that the significance and direction of the *DID* coefficient remain consistent with the main findings, even after accounting for regional factors.

Additionally, we consider that SOEs providing HPWE may relate to employee salary levels, workforce size, and other performance-influencing factors. Thus, we control for salary level (*Salary*), workforce size (*Labor*), and the number of retirees (*Retire*). Salary level is total employee compensation divided by the number of employees, workforce size is the natural logarithm of total employees, and retiree count is the natural logarithm of retired employees. Regression results in column (2) of Table 5 indicate a significantly positive *DID* coefficient at the 1% level, aligning with primary findings.

Change control group

To address potential systematic differences between SOEs providing HPWE and those that do not, we employ the Propensity Score Matching (PSM) method. Initially, using a logit model, we match treatment and non-HPWE SOEs based on control variables consistent with model (1). Employing a 1:1 matching ratio, we then regress model (1). Results in column (1) of Table 6 indicate a significantly positive coefficient for *DID*. Subsequently, considering the policy background across provinces and SOE workforce sizes as influential factors, we construct control groups based on labour scales within each province. Non-HPWE SOEs with similar labour scales are selected as controls. Regression results in column (2) of Table 6 affirm a significantly positive coefficient for *DID* across all cases, aligning with our primary findings.

Table 4. Robustness test: adjust research sample.

	(1) <i>Froa</i>	(2) <i>Froa</i>
<i>DID</i>	0.0113*** (3.5506)	0.0118*** (3.0941)
Controls	Yes	Yes
Year FE	Yes	Yes
Firm FE	Yes	Yes
Observations	11897	11438
Adj.R ²	0.4685	0.4718

Based on the principle of “transfer first, and then reform” mentioned in the “guiding opinions”, we used the time when SOEs disclosed the information on the HPWE as the specific time for SOEs to transfer the HPWE-related assets and then set the independent variable (*DID*). (Unless otherwise specified, “Controls” represent the control variables in column 2 of Table 3.).

Table 5. Robustness test: add the control variables.

	(1) <i>Froa</i>	(2) <i>Froa</i>
<i>DID</i>	0.0086*** (3.4193)	0.0079*** (3.2120)
<i>Public</i>	0.0105 (0.4202)	
<i>Market</i>	0.0017 (1.1991)	
<i>Salary</i>		0.0031 (1.2272)
<i>Labor</i>		−0.0003 (−0.1171)
<i>Retire</i>		0.0000 (0.1350)
Controls	Yes	Yes
Year FE	Yes	Yes
Firm FE	Yes	Yes
Province FE	Yes	No
Observations	12134	12125
Adj.R ²	0.4649	0.4644

This table presents the results of adding other control variables, column (1) and column (2) add control variables at the regional and firm levels, respectively.

Change dependent variable

In order to improve robustness of the regression results, we further change the dependent variable in model (1). Among them, columns (1) to (2) of Table 7 are the regression results using the return on assets of period $t + 2$ and period $t + 3$ as dependent variable; column (3) is the regression results using EBIT of period $t + 1$ as dependent variable; column (4) is the regression result using the return on net assets of period $t + 1$ as dependent variable; column (5) is the regression result using operating profit of period $t + 1$ divided by total assets as dependent variable; column (6) is the regression result using net profit margin on sales of period $t + 1$ as dependent variable. From the results of each column in Table 7, it can be found that the coefficients of *DID* are all significantly positive, which is consistent with the main result.

Table 6. Robustness test: change control group.

	(1) <i>Froa</i>	(2) <i>Froa</i>
<i>DID</i>	0.0069* (1.9281)	0.0072*** (2.6291)
Controls	Yes	Yes
Year FE	Yes	Yes
Firm FE	Yes	Yes
Observations	3246	4472
Adj.R ²	0.4668	0.4518

This table presents the results of reselecting the control group using the PSM method. Column (1) selects enterprises with the closest PSM score calculated based on the control variables, and column (2) selects enterprises of the closest labour scales in the same province.

V. Further analysis

Mechanism analysis

Through the above empirical results, the HPWE reform has significantly improved the performance of SOEs. According to the theoretical analysis, we further examine the three mechanisms of the HPWE reform affecting the performance of SOEs. Following Baron and Kenny (1986), we use the following model to test the mechanism:

$$Mediator_{i,t+1} = \beta_0 + \beta_1 DID_{i,t} + \Sigma Controls + \Sigma Year + \delta_i + \varepsilon_{i,t} \quad (3)$$

$$Froa_{i,t+1} = \beta_0 + \beta_1 DID_{i,t} + \beta_2 Mediator_{i,t+1} + \Sigma Controls + \Sigma Year + \delta_i + \varepsilon_{i,t} \quad (4)$$

The dependent variable in model (2) are mediators which are the excess employees, labour productivity and agency costs. Model (2) tests the impact of the HPWE reform on the excess employees, labour productivity, and agency costs. If the coefficient of *DID* in model (2) is significant, then the mediator will be added into model (3). If the coefficient of *Mediator* is significant in model (3), it indicates that the HPWE reform has improved the performance of SOEs through this mechanism.

Reducing excess employees

Referring to the methodology of Johansson et al. (2017), we use model (5) to calculate the excess employees of enterprises.

$$Lab_{i,t} = \beta_0 + \beta_1 Size_{i,t} + \beta_2 Ppe_{i,t} + \beta_3 Growth_{i,t} + \Sigma Year + \Sigma Industry + \varepsilon_{i,t} \quad (5)$$

In model (5), *Lab* indicates the size of employees, which is equal to the number of employees divided by the total assets and multiplied by 1,000,000; *Size* indicates the firm size, which is equal to the logarithm of the total assets; *Ppe* indicates the proportion of fixed assets, which is equal to the net fixed assets divided by the total assets; *Growth* indicates the growth rate of assets, which is equal to the growth rate of the total assets; model (5) also controls year fixed effect and industry fixed effect. Each coefficient in model (5) is obtained through regression, and the excess employee (*OverLab*) is equal to the actual number of employees minus the expected number of employees, that is, the residual ($\varepsilon_{i,t}$) in model (5). Then, we use *Overlab* as dependent variable in model (3) and as mediator in model (4). The results of mechanism test are shown in Table 8.

In column (1) of Table 8, it can be found that the coefficient of *DID* is significantly negative at the 5% level, indicating that the HPWE reform has significantly reduced the excess employment of SOEs. Column (2) of Table 8 presents the regression result after adding *DID* and *OverLab* to model (3). It can be found that the coefficient of *DID* is significantly positive at the 1% level, while the coefficient of *OverLab* is significantly negative at the 1% level, indicating that the excess employee of SOEs plays a partial mediating role. The above results demonstrate that the HPWE reform has significantly reduced the excess employees of SOEs and thus improved the performance of SOEs.

Improving labor productivity

In Table 9, labour productivity (*Labeff*) is measured as the natural logarithm of output per

Table 7. Robustness test: change dependent variable.

	(1) <i>F2roa</i>	(2) <i>F3roa</i>	(3) <i>Febit</i>	(4) <i>Froe</i>	(5) <i>Froam</i>	(6) <i>Fnpm</i>
<i>DID</i>	0.0121*** (3.5262)	0.0118*** (3.1640)	0.0071*** (2.6570)	0.0189*** (2.1042)	0.0075*** (2.7983)	0.0169*** (2.0731)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	10757	9551	12135	12134	12135	12134
Adj.R ²	0.4515	0.4378	0.4911	0.2556	0.5391	0.3978

In order to further improve robustness of the regression results, we change the dependent variable. Columns (1) to (2) present regression results with the return on assets of period t+2 and t+3 as dependent variables. Columns (3) to (6) shows the regression results using EBIT of period t+1, the return on equity of period t+1, operating profit of period t+1 divided by total assets, net profit margin on sales of period t+1 as dependent variables.

employee, following Kale, Ryan, and Wang (2019). Model (3) uses labour productivity as the dependent variable, while model (4) employs it as a mediator. Column (1) of Table 9 demonstrates a significant positive effect of the HPWE reform on labour productivity, with DID coefficient significantly positive at the 1% level. In column (2), both *DID* and *Labeff* coefficients are significantly positive at the 5% and 1% levels, respectively, indicating a partial mediation effect. Thus, the HPWE reform not only boosted SOEs' labour productivity but also contributed to their overall performance enhancement.

Alleviating agency costs

Referring to Ang, Cole, and Lin (2000), we use the sum of sales and management expenses divided by operating income to measure agency costs (*Agency*), and *Agency* is used as the dependent variable in models (3) and as a mediator in model (4). The regression results are shown in columns (1) and (2) of Table 10. Additionally, to directly test the effect of the HPWE reform on the salary incentives of SOEs, we use model (6) to examine the impact of the HPWE reform on the performance sensitivity of executive compensation.

$$\begin{aligned} \text{Managesalary}_{i,t} = & \beta_0 + \beta_1 \text{DID}_{i,t} + \beta_2 \text{DID}_{i,t} \\ & * \text{Roai}_{i,t} + \Sigma \text{Controls} + \Sigma \text{Year} \\ & + \delta_i + \varepsilon_{i,t} \end{aligned} \quad (6)$$

In model (6), *Managesalary* is the logarithm of the total compensation of executives, and the other control variables and fixed effects are consistent

Table 9. Mechanism test: improving labour productivity.

	(1) <i>Labeff</i>	(2) <i>Froa</i>
<i>DID</i>	0.1076*** (3.2235)	0.0052** (2.1477)
<i>Labeff</i>		0.0272*** (14.0171)
Controls	Yes	Yes
Year FE	Yes	Yes
Firm FE	Yes	Yes
Observations	12132	12132
Adj.R ²	0.8205	0.4971

The table presents the mechanism test of improving labour productivity. The results in column (1) indicate that the HPWE reform has improved the labour productivity of enterprises, the results in column (2) prove that the labour productivity has a significant positive impact on enterprise performance.

with model (1). The intersection of *DID* and *Roa* is added to model (6). If the HPWE reform improves the performance sensitivity of executive compensation, then the coefficient of *DID***Roa* will be significantly positive. The regression results are shown in column (3) of Table 10.

Column (1) of Table 10 displays the regression results of the HPWE reform on agency costs. The coefficient of *DID* is significantly negative at the 5% level, indicating that the HPWE reform has led to a significant reduction in the agency cost of SOEs. In column (2) of Table 10, the coefficient of *DID* is significantly positive at the 1% level, while the coefficient of *Agency* is significantly negative at the 1% level, indicating that agency costs play a partial mediating role. Column (3) of Table 10 presents the regression result of model (6). The coefficient of *DID***Roa* is significantly positive at the 10% level. This finding indicates that the HPWE reform has significantly increased the performance sensitivity of executive compensation in SOEs, further demonstrating that the HPWE reform has alleviated agency costs.

Table 8. Mechanism test: reducing excess employees.

	(1) <i>OverLab</i>	(2) <i>Froa</i>
<i>DID</i>	-0.1154** (-2.1164)	0.0071*** (2.9083)
<i>OverLab</i>		-0.0091*** (-6.0604)
Controls	Yes	Yes
Year FE	Yes	Yes
Firm FE	Yes	Yes
Observations	12132	12132
Adj.R ²	0.6684	0.4749

The table presents the mechanism test of reducing excess employees. The results in column (1) shows that the HPWE reform has significantly reduced the excess employment of SOEs, and the column (2) proves that excess employees have a significant negative impact on corporate performance.

Cross-sectional tests

To strengthen our findings, we conduct cross-sectional tests focusing on labour density and ownership-control separation. SOEs face different pressures regarding HPWE reform. Those under greater pressure typically have more economic resources tied up in HPWE and require more employees. By separating HPWE, these SOEs can cut excess staff and redirect resources to core operations, leading to a more pronounced positive impact on business performance.

Table 10. Mechanism test: alleviating agency costs.

	(1) <i>Agency</i>	(2) <i>Froa</i>	(3) <i>Managesalary</i>
<i>DID</i>	−0.0077** (−2.2875)	0.0089*** (3.4711)	−0.0936** (−2.4554)
<i>Agency</i>		−0.0471*** (−3.3280)	
<i>DID*RoA</i>			0.8990* (1.7351)
<i>RoA</i>	−0.2749*** (−10.0151)	0.0717*** (2.7056)	0.9453*** (7.4236)
Controls	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Observations	11946	11946	12101
Adj.R ²	0.7861	0.4667	0.8013

The table presents the mechanism test of alleviating agency costs. The results in columns (1) and (2) indicate that agency costs play a partial mediating role and the results in columns (3) indicate that the HPWE reform has significantly increased the performance sensitivity of executive compensation in SOEs, further demonstrating that the HPWE reform has alleviated agency costs. ("Controls" here represent the control variables in column 2 of Table 3, excluding *RoA*).

Labour-intensive SOEs, which rely heavily on labour and bear more HPWE responsibilities, are assessed based on total employee compensation as a percentage of operating income, classified into high and low labour intensity groups (Dewenter and Malatesta 2001). Regression analyses using model (1) reveal a significantly positive *DID* coefficient in the high labour intensity group (Table 11, columns (1) and (2)). In contrast, the low labour intensity group's *DID* coefficient is positive but not significant. An inter-group coefficient difference test shows significant disparities between the two *DID* coefficients at the 5% level. These findings highlight the stronger impact of HPWE reform on high labour intensity SOEs.

SOEs typically have a hierarchical equity control chain, leading to an ownership structure where control exceeds ownership. This extended control chain

increases the risk of controlling shareholders infringing on smaller shareholders' rights and complicates oversight and incentive mechanisms, resulting in higher agency costs (Lemmon and Lins 2003; Pagano and Volpin 2005). Consequently, the agency problem is more pronounced in SOEs with significant ownership-control separation, allowing managers to exploit their positions. Separating HPWE encourages managerial focus on core operations, enhances the effectiveness of salary incentives, and produces a more pronounced marginal effect on performance. We stratify samples into high and low separation groups based on ownership-control separation and employ regression analysis using model (1). Results in Table 12, columns (1) and (2), reveal a significantly positive *DID* coefficient at the 1% level in the high separation group, while the low separation group shows a positive but insignificant coefficient. Additionally, an inter-group coefficient difference test indicates significant disparities between the two groups at the 5% level. These findings highlight the stronger impact of HPWE reform on SOE performance in entities with higher ownership-control separation.

Table 11. Cross-sectional tests: labour intensity.

	(1) Low labour intensity <i>Froa</i>	(2) High labour intensity <i>Froa</i>
<i>DID</i>	0.0016 (0.4778)	0.0129*** (3.0228)
Controls	Yes	Yes
Year FE	Yes	Yes
Firm FE	Yes	Yes
Observations	6109	5829
Adj.R ²	0.5363	0.4852
<i>Chi2</i>		4.523**
<i>Pvalue</i>		0.0334

The table presents the heterogeneity test results of the corporate performance in different subsamples. Columns (1) and (2) report the results conditional on the samples below- versus above- the median of the mean labour intensity, respectively.

VI. Conclusions and policy implications

Conclusion

SOEs play a crucial role in China's socialist market economy, but they also face issues of inefficiency. This article examines the significant impact of

Table 12. Cross-sectional tests: separation of ownership and control.

	(1) Low degree of separation <i>Froa</i>	(2) High degree of separation <i>Froa</i>
<i>DID</i>	0.0037 (1.0810)	0.0142*** (3.4276)
Controls	Yes	Yes
Year FE	Yes	Yes
Firm FE	Yes	Yes
Observations	7572	4371
Adj.R ²	0.5065	0.4392
<i>Chi2</i>		4.098**
<i>Pvalue</i>		0.0429

The table presents the heterogeneity test results of the corporate performance in different subsamples. Columns (1) and (2) report the results conditional on the samples below- versus above- the median of the mean separation of ownership and control respectively.

policy burden divestiture on SOE performance, using the separation of HPWE services in China as a quasi-natural experiment to alleviate potential endogeneity issues. The study demonstrates that this reform significantly enhances the performance of Chinese SOEs through three mechanisms: reducing redundant labour, increasing productivity, and lowering agency costs. Furthermore, performance improvements are particularly pronounced in labour-intensive enterprises and those SOEs with a greater separation between ownership and control.

This study's findings are consistent with the existing literature on the inefficiency of SOEs, particularly in socialist and emerging economies where political interventions often impede operational efficiency. Although privatization has been proposed as a potential solution, this study highlights the value of focusing on the separation of policy burdens as an alternative approach. The results demonstrate that alleviating these burdens can significantly enhance SOE efficiency, thereby contributing to the literature on SOE reform and providing practical insights for policymakers. Future reforms should prioritize the divestment of non-core functions, enabling SOEs to concentrate on their primary operations and thereby improve their overall economic contributions.

Policy implications

This study provides key policy implications for reforming SOEs. First, SOEs should divest non-core functions to reduce government intervention

and alleviate policy burdens, as evidence indicates that separating social functions enhances operational performance and promotes fair market competition. By concentrating on core operations, SOEs can boost overall efficiency. Second, following divestment, SOEs must strengthen incentive systems for executives and employees to enhance motivation and accountability, positively impacting productivity and compensation effectiveness. Reinvesting savings from divestment into improved incentive structures can lower agency costs and enhance performance. Lastly, SOEs should streamline control chains and minimize the separation between ownership and control. While complex ownership structures may limit government interference, they can weaken internal oversight and exacerbate agency issues. Simplifying control chains enables SOEs to focus on core operations and enhance management oversight, leading to improved performance.

Additionally, this study has broader implications for SOE reform in other countries. Given their vital role in economic development and resource management, governments should prioritize alleviating policy burdens by reducing non-commercial obligations. Implementing reforms like China's HPWE separation can enhance SOE efficiency by enabling them to concentrate on core functions, thereby minimizing agency costs associated with conflicting objectives. It is also essential for governments to recognize the risks of excessive social responsibilities that hinder market growth and inflate costs. Countries facing similar inefficiencies, whether developing or developed, could benefit from these strategies. By adopting these approaches, policymakers can optimize SOE performance and promote sustainable economic growth.

Limitations and future research scope

Although this article explores the mechanism by which the separation of the HPWE improves the operational performance of state-owned enterprises, there may still be other unidentified mechanisms or pathways that require further in-depth research. However, the robustness of our study will not be affected by research on other mechanisms. Future studies can further explore other unidentified mechanisms or pathways

through which the separation of the policy burdens enhances the operational performance of state-owned enterprises. In addition to focusing on economic benefits, attempts can also be made to comprehensively assess the impact of the HPWE reform on SOEs and even the entire society from multiple dimensions, including social effects and environmental effects.

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